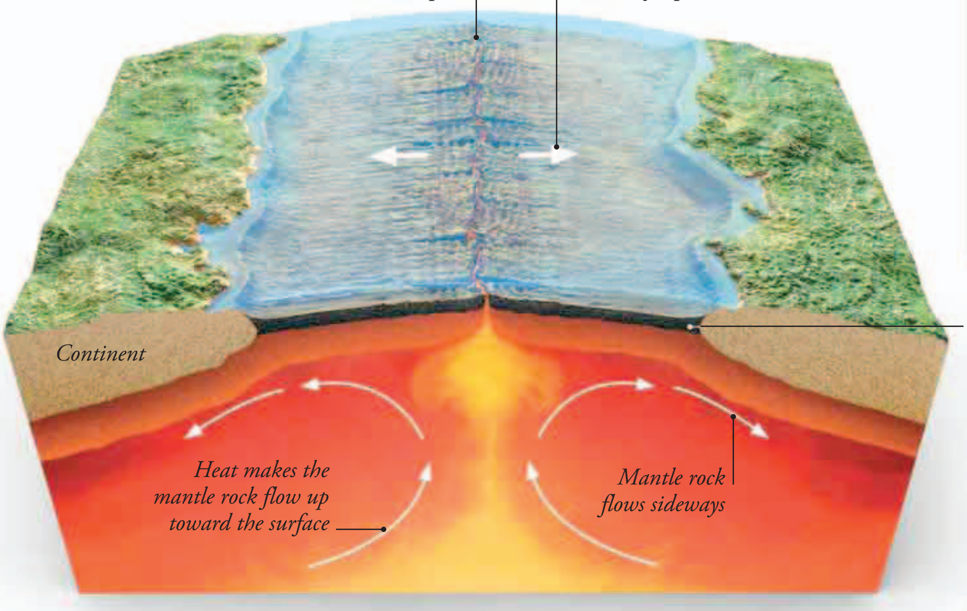
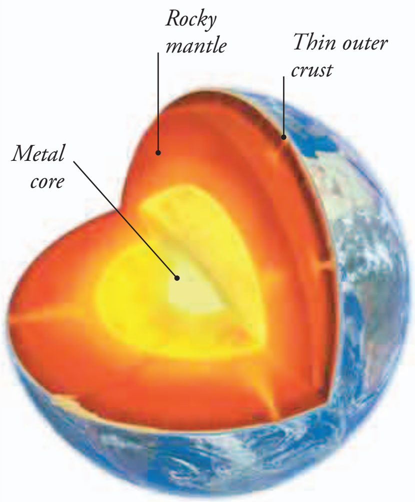
How oceans formed

An ocean is not just a big hollow filled with salty water. Its floor is made of a special type of rock that forms in the gaps where Earth's crust has been dragged apart by forces within the planet. This rock is the cool, brittle shell of the deep, hot mantle that lies below. The continents are thicker slabs of lighter rock that float on the mantle like rafts.



LAYERED EARTH



Earth was created about 4.6 billion years ago when dust and rocks orbiting the Sun began to clump together. As the planet grew, it attracted iron-rich meteorites, which slammed into Earth, melting on impact. The heat built up until the whole mass of rock and metal melted. The heavier metals, such as iron, then sank toward the center of the planet to form a hot metallic core surrounded by the thick rocky mantle and cool outer crust.

MOVING PLATES

Nuclear energy deep inside the planet keeps the rocky mantle very hot. High pressure stops the rock from melting, but the heat makes it soft enough to flow in currents that rise very slowly, flow sideways near the surface, then sink. As the mantle rock flows sideways, it drags the brittle crust with it. This breaks the crust into many separate, moving plates, which carry the continents. New ocean floor forms where these plates are pulling apart.

FLOATING ROCK

The mantle is made of a very heavy rock called peridotite. A slightly lighter rock called basalt forms the oceanic crust—the bedrock of the ocean floors. The continents are made of granite and similar rocks, which are even lighter than basalt. This enables the continents to float on the heavy mantle, like ice on water, and is one reason why the continents rise above the ocean floors.

Peridotite

Basalt

Granite